ALYSSA PLAYER

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PROFESSIONAL SUMMARY

MS in Data Science candidate with 5+ years of research experience building machine learning models and statistical analyses in Python and R. Skilled in translating complex datasets into actionable insights, with hands-on experience in customer behavior modeling, data visualization, and end-to-end ML pipelines.

EDUCATION

Master of Science, Data Science **August 2025 – Present**
Boston University, GPA 3.9 – Graduation Date: December 2026

Bachelor of Arts, Biology **August 2020 – May 2024**
Occidental College, Honors in Biology for Independent Research Thesis

Google Data Analytics, Professional Certificate **October 2024 – August 2025**

WORK EXPERIENCE

California Institute of Technology
Research Technician, Beacon Center for Single Cell Biology / Björkman Lab — July 2024 – Present

- Execute high-throughput antigen reactivity screening (~20k samples/run) on the Beacon Opto-Fluidic System, contributing to discovery of novel therapeutic candidates
- Build and manage JavaScript-automated data-entry templates in Google Sheets/Excel, saving a 3-person team ~3 hours per session and substantially reducing data entry errors
- Produce data visualizations and scientific graphics for manuscripts, academic posters, and the Beacon Center web platform using GraphPad Prism, BioLegend tools, Adobe Illustrator, Photoshop, and PowerPoint
- Serve as primary client contact, facilitating biweekly meetings and translating findings into actionable project updates

Vantuna Research Group, Occidental College
Research Assistant — February 2022 – May 2024

- Completed an independent Honors Thesis applying GLMMs and BACIP models in R to analyze climate-threatened reef species across the Southern California Bight
- Processed and performed QA/QC on a 20-year longitudinal dataset for government-funded reef monitoring projects and annual performance reports while mentoring 5-10 new students per semester in experimental design and data collection

PROJECTS

Application of ML Techniques
MS in Data Science Capstone Project

- Applied gradient boosting and KNN to model customer review scores (1-5) on Olist e-commerce data, reframing a logistics dataset into a marketing analytics tool for identifying satisfaction drivers
- Predicted profit outcomes on 47k+ US e-commerce records via regression and classification pipelines, achieving R^2 of 0.9

Automated Property Valuation
Supervised ML Pipeline | Python, Scikit-learn, Gradient Boosting, Decision Trees

- Predicted residential property values from 77k+ Zillow records; Gradient Boosting achieved a test MAE of $195,927, outperforming Lasso and Decision Tree baselines

PUBLICATIONS

Malecek, K. E., Alvarez, J., Player, A., Keeffe, J. R., Gnanapragasam, P. N. P., Segovia, L. N., Tan, T. K, Schimanski, L & Bjorkman, P. J. (2026). A mosaic RBD-nanoparticle pan-sarbecovirus vaccine elicits nanobodies that target highly conserved epitopes. [Manuscript in preparation]. Biology and Biological Engineering Division, California Institute of Technology.

Player, A., Mũnoz-Williams, C.M., Williams, J.P., Pondella, D.J (2027). Artificial reef models provide refuge habitat to keystone echinoderms against climate stressors. [Manuscript in preparation].

SKILLS

ML & Statistics: Scikit-learn, Regularised Regression, Ensemble Methods, Classification, Clustering, Hyperparameter Tuning
Languages & Tools: Python, R, SQL, JavaScript, HTML, Git/GitHub, VS Code
Data Visualization: Seaborn, Matplotlib, ggplot2, Tableau, Adobe Illustrator
Cloud & Data: MS Azure Data Factory / Synapse, BigQuery, Google Sheets, Excel

VOLUNTEER

Caltech Green Labs · Letters to a Pre-Scientist Program